

Management Information Systems

11. Designing Output and Input

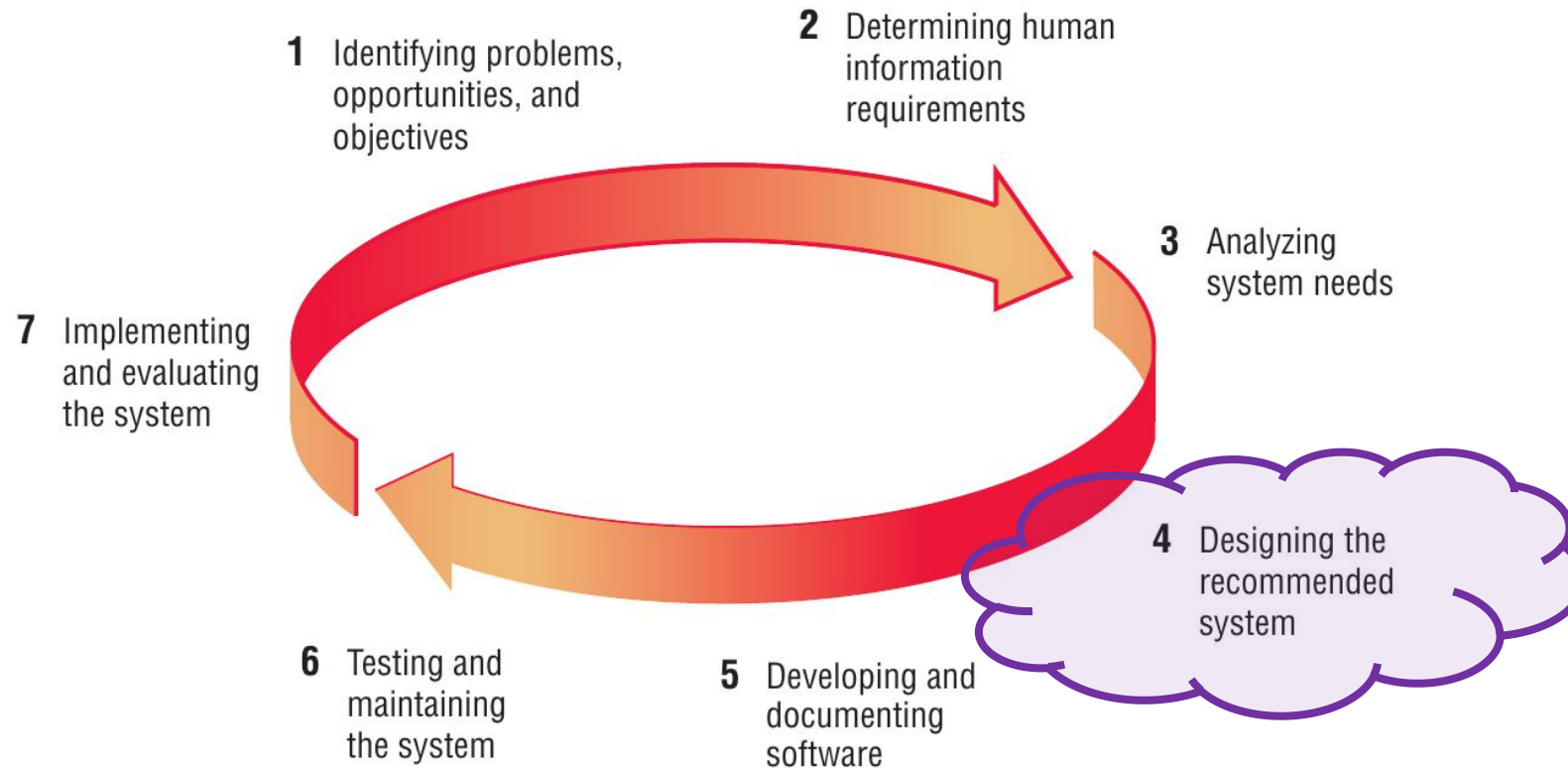
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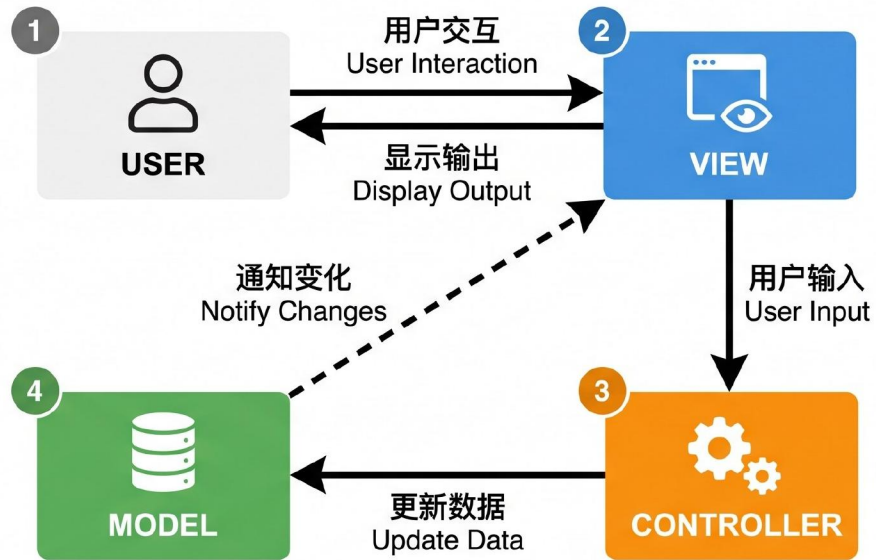


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Tsinghua SEM

Designing



A General Architecture (the MVC Pattern)



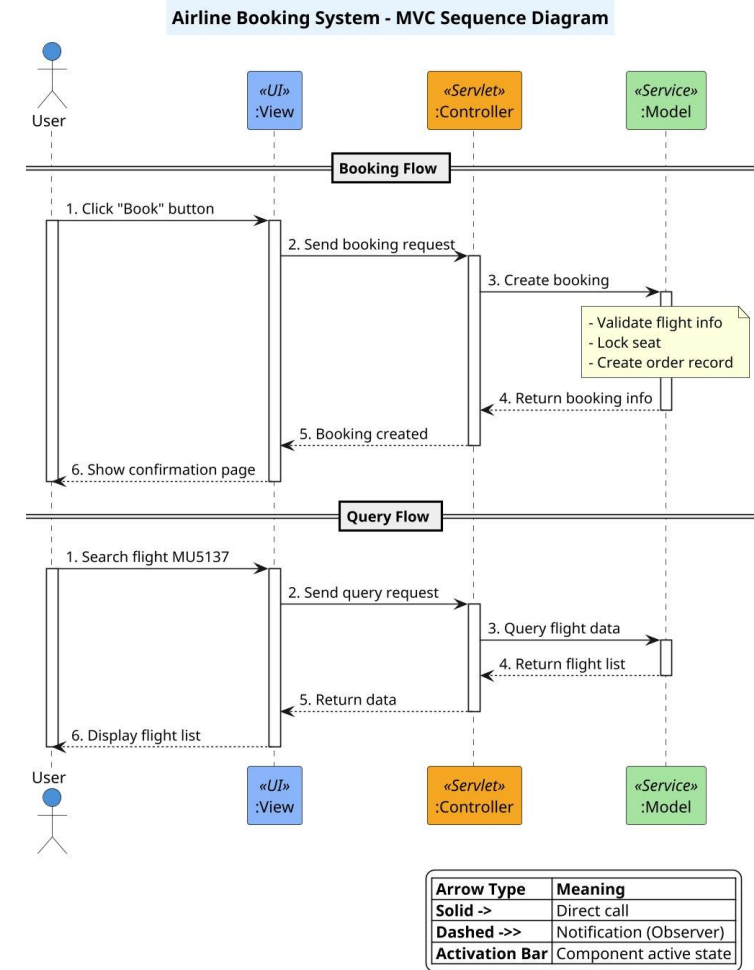
航空订票示例

Step 1: 用户点击"预订" → View

Step 2: View → Controller (发送预订请求)

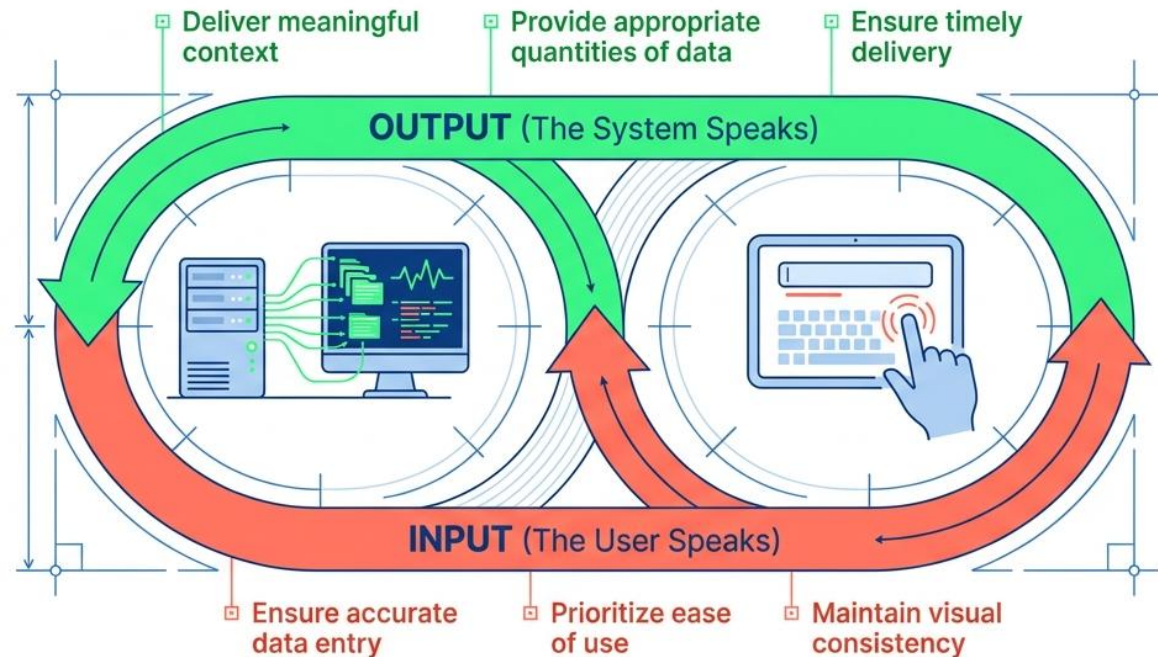
Step 3: Controller → Model (创建订单)

Step 4: Model → View (更新座位状态)



Output and Input

- output and input are two sides of one coin
 - every output can become input, every input shapes output



Core Principle: Right information, right person, right time, right medium.

Designing Output and Input

- Designing Effective Output (Textbook, Chapter 11)
 - Choosing an output method
 - Report design
 - Screen design
 - Web site design
 - Smartphone and tablet design
- Designing Effective Input (Textbook, Chapter 12)
 - Form design
 - Display and Web form design
 - Website design



Designing Effective Output



Output Rules

- Choose the right output method (medium)
- Keep it simple and uncluttered
- Neutralize output bias
- Provide context alongside summarization



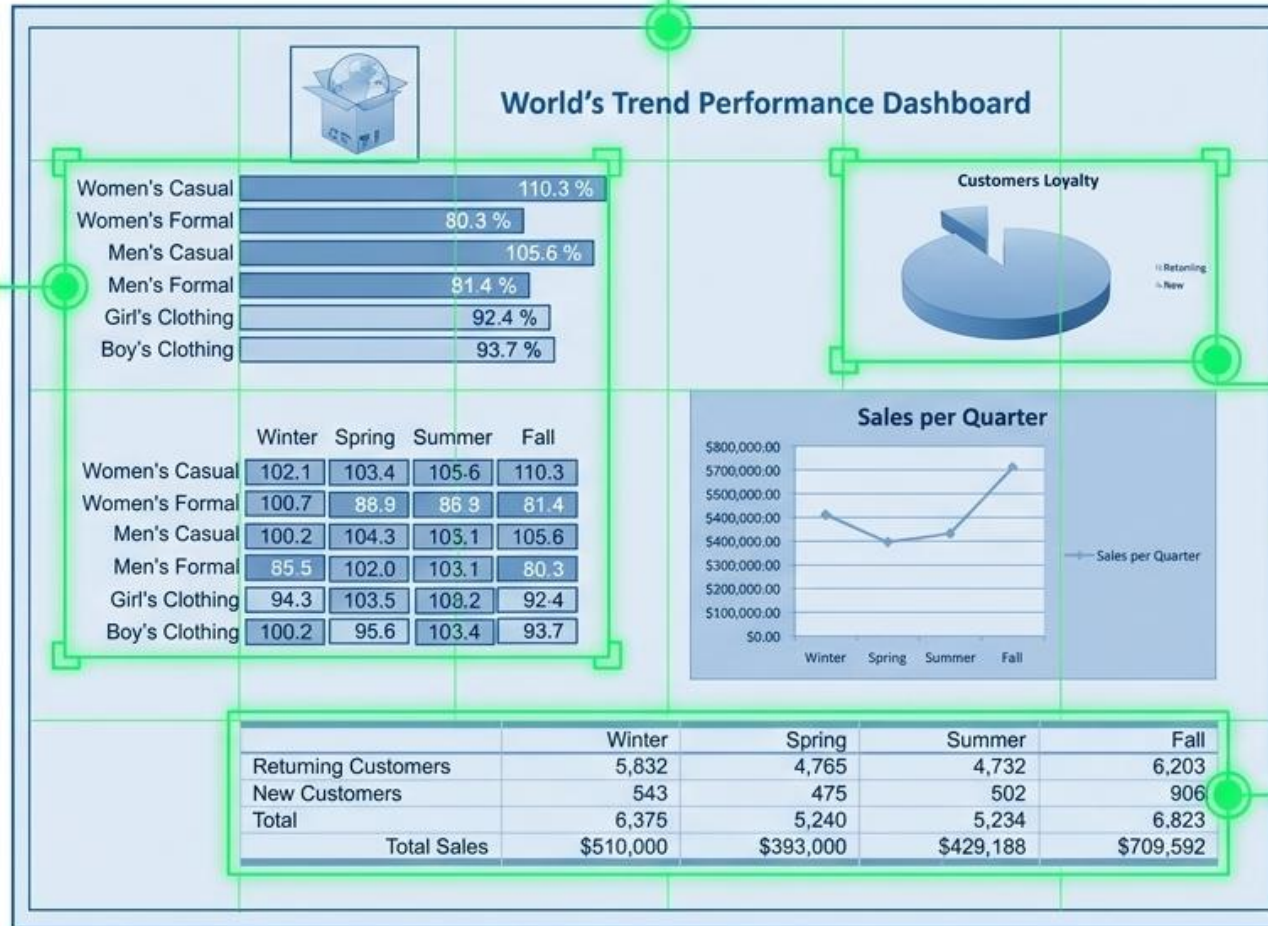
Choosing the Right Output Method (Medium)

	Medium	Advantage	Disadvantage	Best For
	Displays / Dashboards	Interactive, real-time, quiet	Requires user focus and screen space	Frequently accessed, dynamic data manipulation
	Audio / Podcasts	Hands-free, transient	Disruptive in shared spaces, limited retention	Individual alerts, transient messages
	Mobile / Tablets	Highly portable, gesture-driven	Small screen constraints, high risk of loss	On-the-go data consumption and quick interactions
	Printers / Hard Copy	Highly reliable, handles large volumes	Environmentally unfriendly, static	Archival records and regulatory compliance

Keep it Simple and Uncluttered

Contextual Framing:

Data is grounded in business context (e.g., trend performance).



Uncluttered Canvas:

The entire dashboard is constrained to a single screen—no scrolling required.

Meaningful Grouping:

Information that logically belongs together is clustered.

Performance Measures:

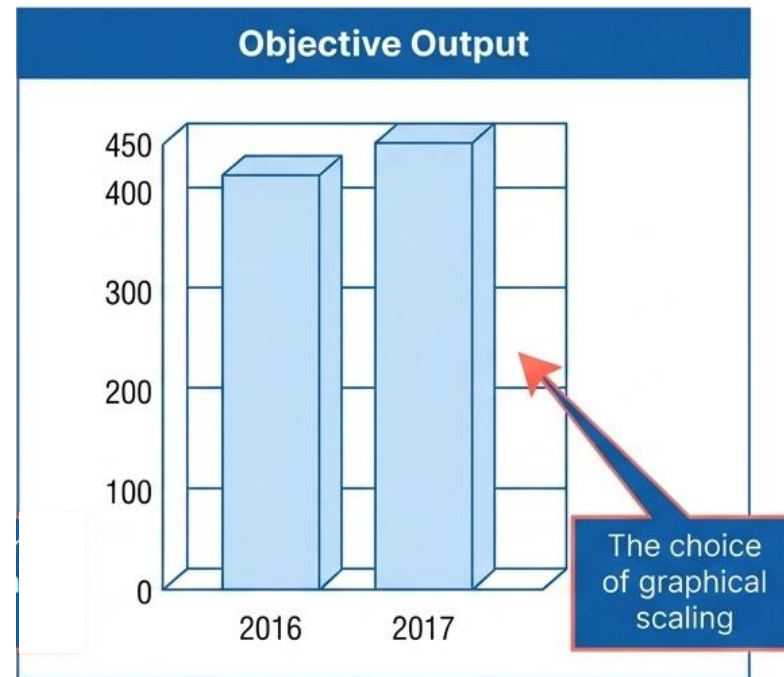
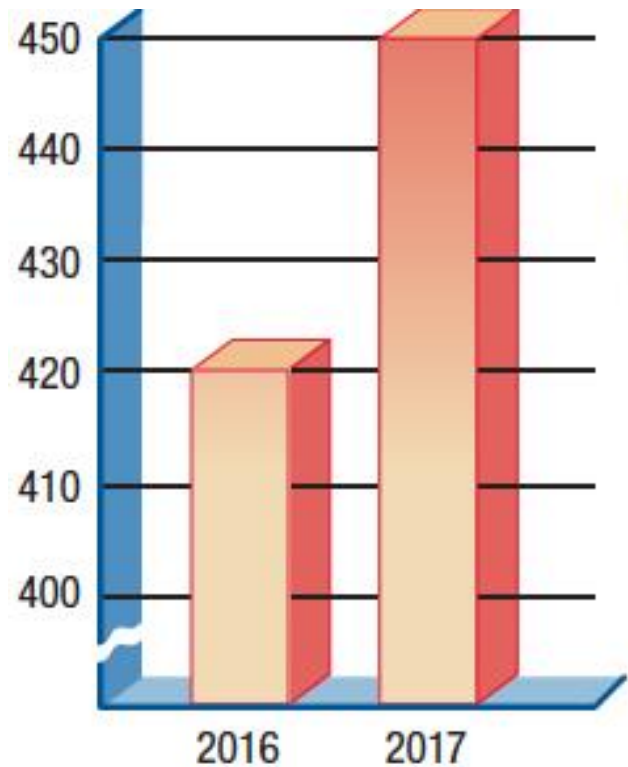
Summarization is prioritized over raw data dumps.

Neutralize Output Bias

- Output is NOT neutral: design choices shape decisions
 - Sorting bias: Alphabetical vs. chronological vs. ranked → different conclusions
 - Graphical bias: Bar chart vs. pie chart vs. table → different impressions
 - Truncation bias: Where you set the Y-axis changes perception
 - Aggregation bias: Summaries hide outliers; averages erase distribution
- Analysts should
 - avoid unnecessarily biasing output
 - make users aware of the possible biases in output



Example: A Misleading Graph



Same Data, Different Decisions?

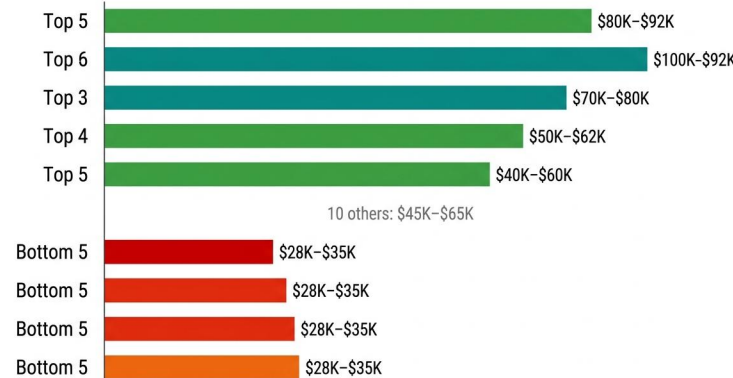
- Imagine you are a regional manager reviewing sales performance.

VIEW A: Complete Sales Performance Ranking

Rank	Name	Region	Monthly Sales (\$)	Target (%)
1	James Davies	North	\$28,450	78%
2	Sarah Green	West	\$29,910	81%
3	Mark Wilson	South	\$31,500	75%
4	Emily Brown	East	\$33,180	84%
5	David Jones	North	\$34,860	80%
6	Laura Lee	West	\$45,300	88%
7	Robert Taylor	South	\$47,120	92%
8	Jessica Evans	East	\$48,950	90%
9	Christopher Miller	North	\$50,770	87%
10	Linda Moore	West	\$52,590	89%
11	Thomas Smith	South	\$54,410	91%
12	Angela White	East	\$56,230	93%
13	Steven Martin	North	\$58,050	86%
14	Patricia Taylor	West	\$59,870	95%
15	Daniel Clark	South	\$61,690	94%
16	Karen Davis	East	\$63,510	97%
17	Joseph Allen	North	\$85,020	102%
18	Susan Roberts	West	\$87,490	105%
19	William Jackson	South	\$89,960	108%
20	Mary Thompson	East	\$92,430	110%

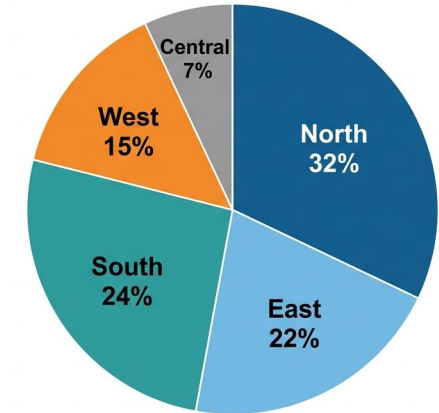
Source: Regional Sales Database, March 2026

VIEW B: Top & Bottom Performers



Source: Regional Sales Database, March 2026

VIEW C: Sales by Region



Source: Regional Sales Database, March 2026

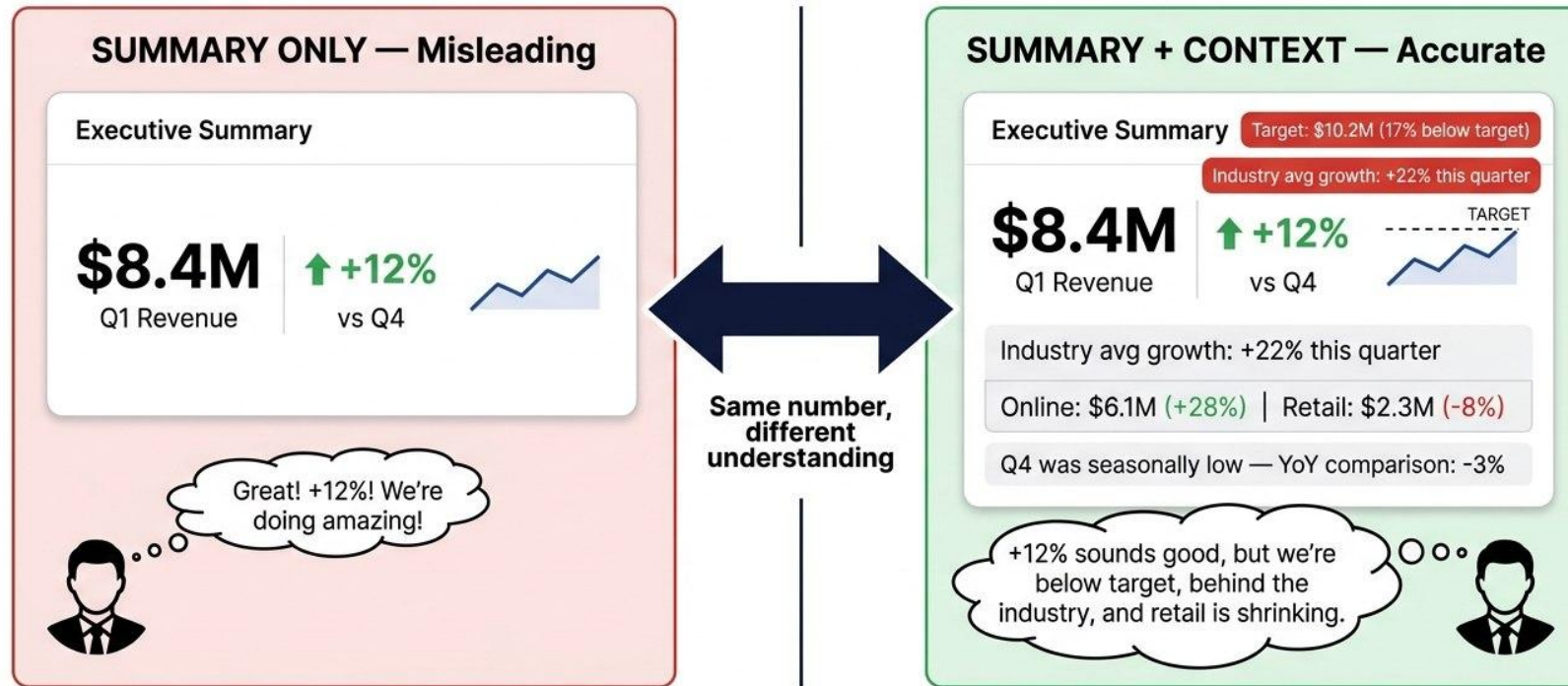
"Let me review everyone carefully"

"Promote the green ones, fire the red ones"

"Invest more in North, figure out what's wrong with Central"

Provide Context alongside Summarization

Summarization Without Context Is Misleading



“A number without context is just a number. Context turns data into information.”

Designing Effective Input



Input Rules

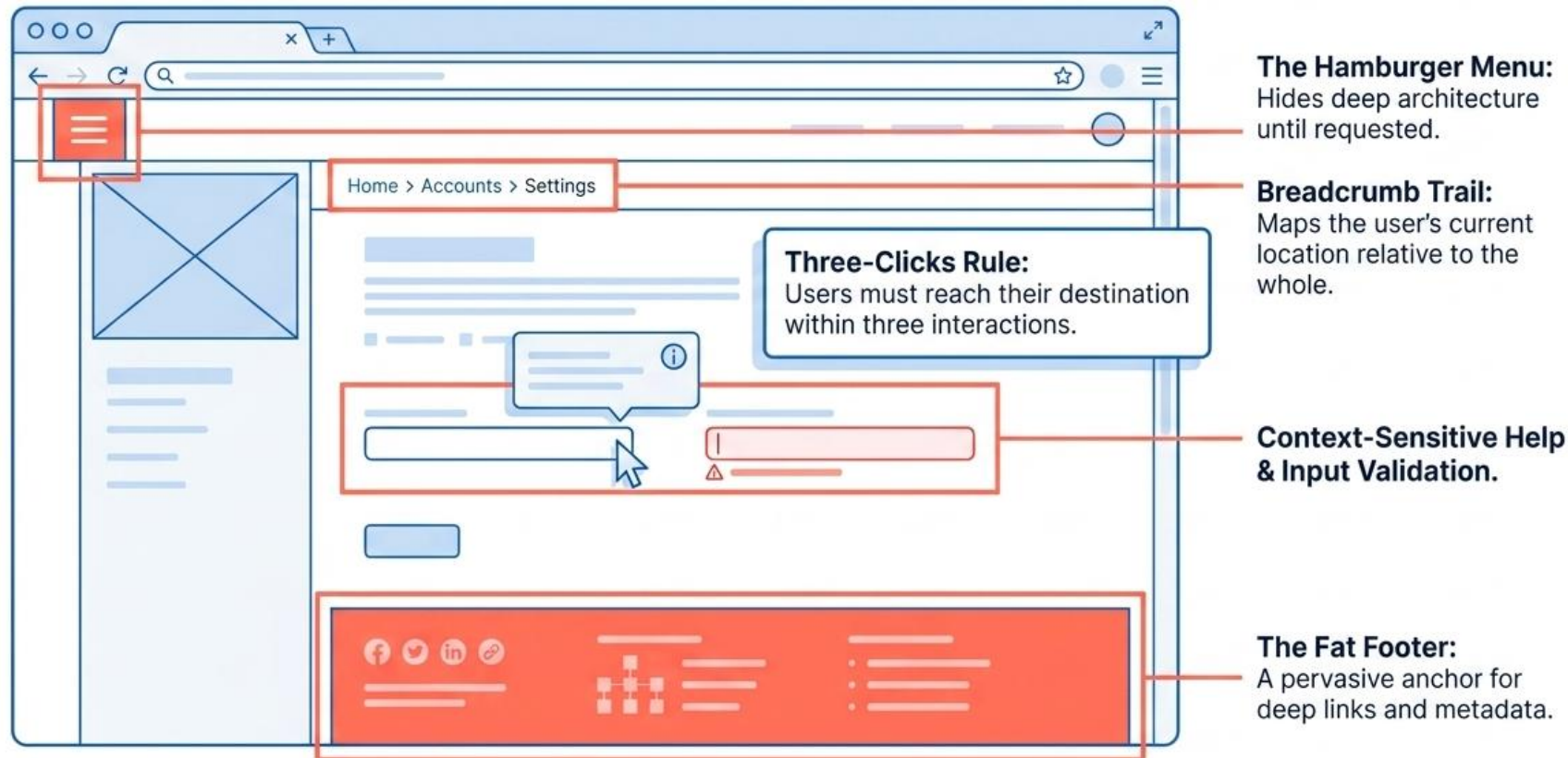
- Choose the right input method (Medium)
- Ensure accurate completion through smart navigation, defaults, and validation
- Follow the natural flow (top-down, left-right)
- Use the proper GUI control for the data type



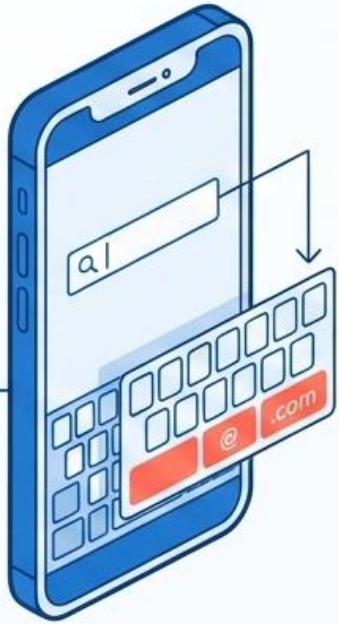
Choosing the Right Input Method (Medium)

Medium		Advantage	Disadvantage	Best For
	Keyboard / Typing	High speed for text entry, universal and familiar, supports complex data formats	Prone to typos and transcription errors, requires desk and focused attention	Long-form data entry, coding, detailed forms, office environments
	Voice / Audio Input	Hands-free operation, natural and fast for simple commands, supports accessibility	Sensitive to ambient noise, privacy concerns in public, limited for structured data	Field work, driving/warehouse contexts, accessibility needs, short commands
	Barcode / QR Code Scanning	Instant and accurate data capture, eliminates manual entry errors, widely supported by hardware	Requires pre-printed codes or labels, limited to encoded data only, scanner dependency	Inventory management, retail POS, warehouse logistics, asset tracking
	Touch / Gesture	Intuitive and learnable, no peripherals needed, supports visual interaction	Finger fatigue on long forms, low precision for detailed data, limited by screen size	Kiosks and POS systems, mobile apps, casual browsing, short selection tasks

Ensure Accurate Completion through Smart Navigation, Defaults and Validation



Smart Defaults: Anticipate User Input



Contextual Keyboards: HTML5 input types (email, tel, url) instantly adapt native keyboards.

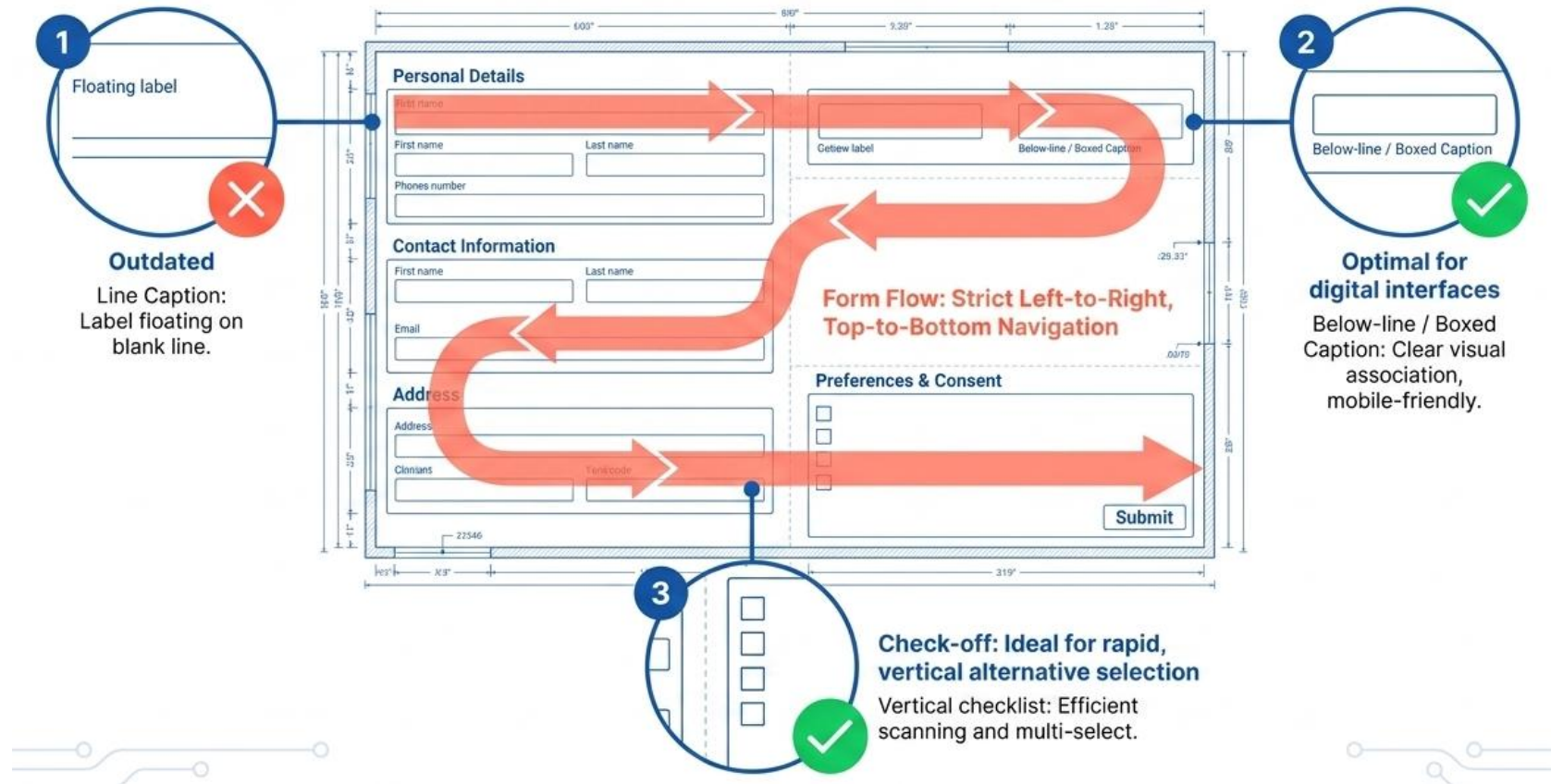


Datalist Autocomplete: Predefined suggestions appear instantly to make entry effortless.



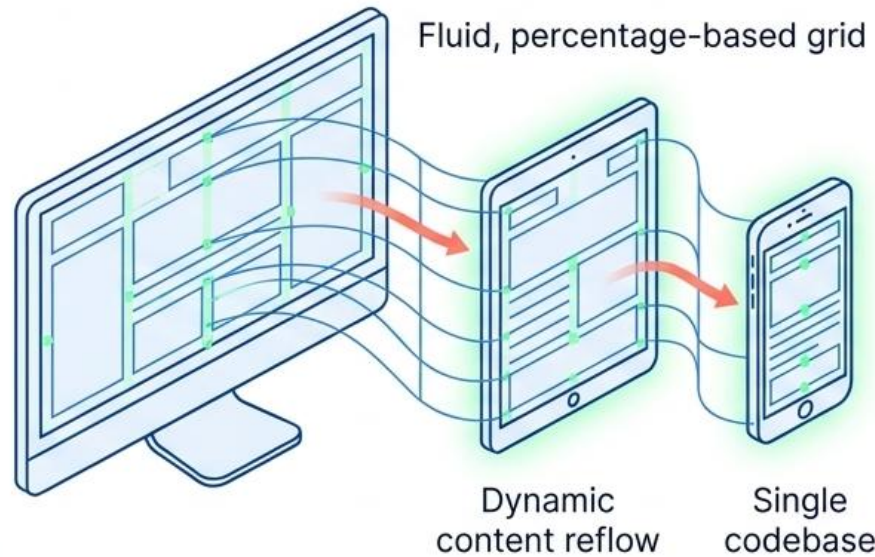
Native Calendars: Error-proof, interactive date-pickers replace manual string typing.

Follow the Natural Flow (Top-Down, Left-Right)

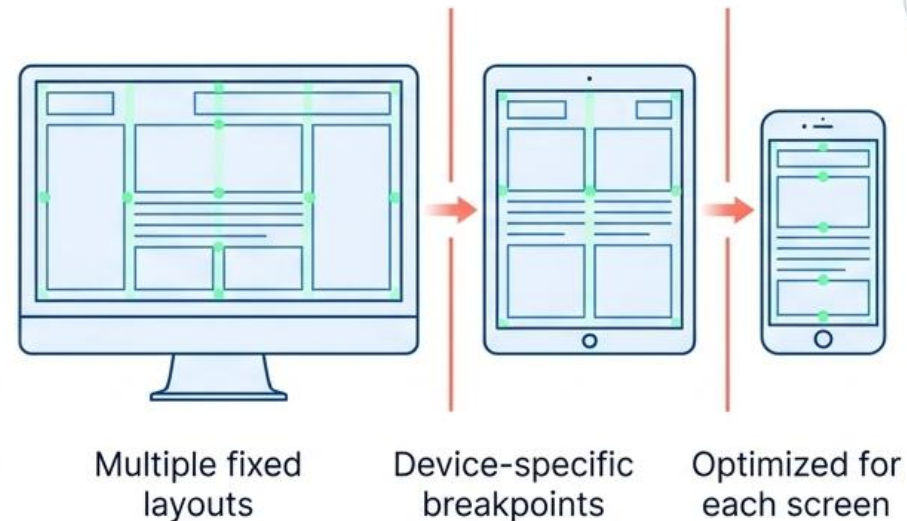


Scaling across Multi-Device Reality

Responsive Design

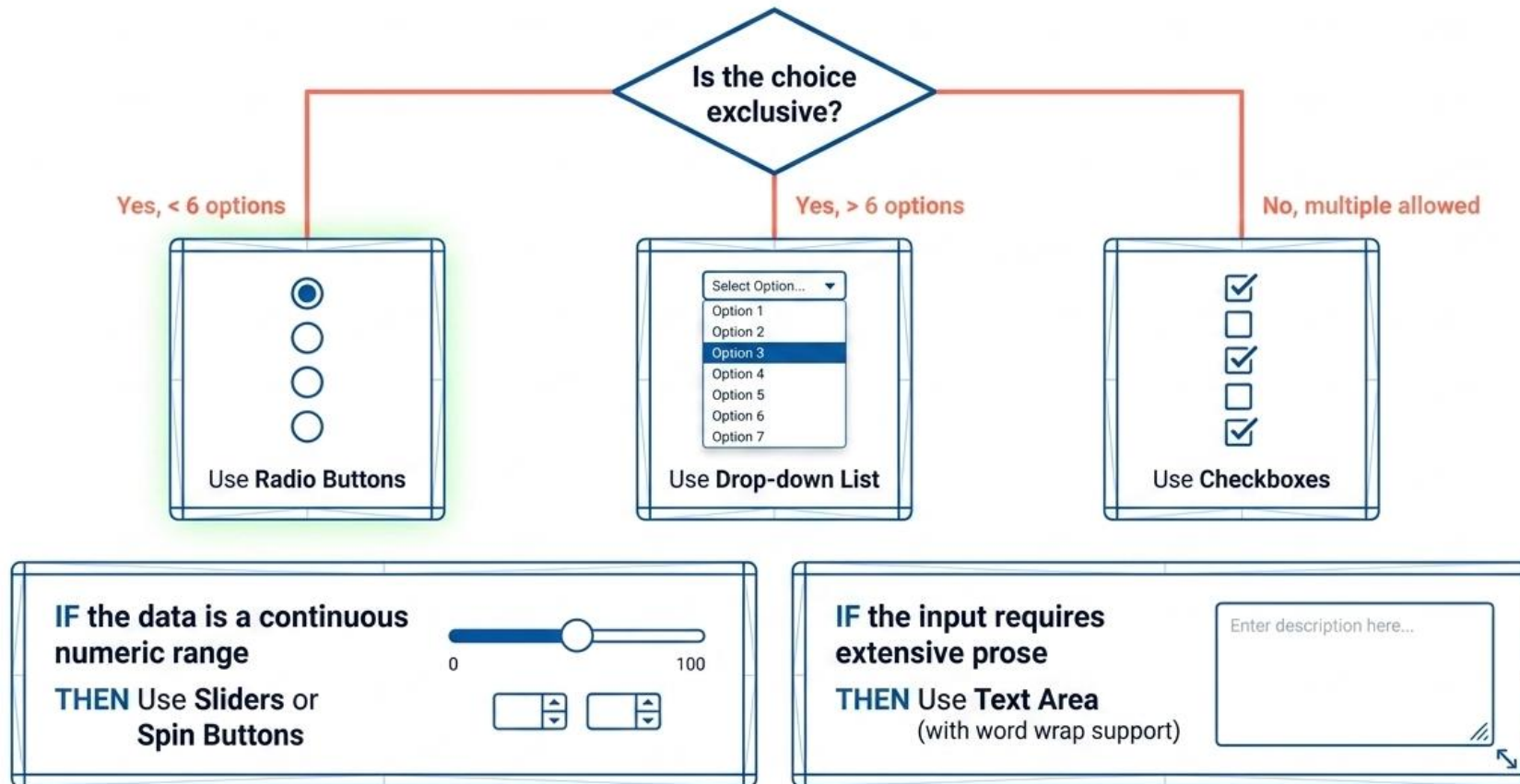


Adaptive Design



Modern multi-device design relies on **Flat Web Design**—clean, **two-dimensional**, and **minimalistic**.

Use the Proper GUI Control for the Data Type



Designing Input and Output

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Input Rules

- Choose the right input method (Medium)
- Ensure accurate completion through smart navigation, defaults, and validation
- Follow the natural flow (top-down, left-right)
- Use the proper GUI control for the data type



Tasks

- Individual tasks
 - Reading: Textbook, Chapters 11-12
 - Answer the review questions (on YuKeTang)
 - Answer the additional review question (on WebLearning)
- Group tasks
 - Refine your prototypes (no submission required)
- Class presentations
 - May 25: Group 3 (王廷轩组, 智云科技) and Group 4 (蓝天翔组, 卢米纳电子)
 - Content: analysis using UML, and refined prototype.
 - Time: 10~20 minutes

